

RYAN LOUIE

CONTACT INFORMATION	Stanford University Dept. of Computer Science http://youralien.github.io	(949) 562-1620 rylouie@stanford.edu
APPOINTMENTS	Stanford University Computer Science , Stanford, CA <i>Postdoc advised by Diyi Yang and Emma Brunskill</i> <i>Supported by Stanford Impact Lab Grant</i>	Sep 2023 - Now
EDUCATION	Northwestern University , Evanston, IL Ph.D. in Technology Social Behavior Committee: Haoqi Zhang (chair), Darren Gergle, Carrie Cai, Andrés Monroy-Hernandez, Jeremy Birnholtz Thesis topic: Human-AI Interface Layers: Enhancing Communication of Intent for AI-Assisted Creative Pursuits and Social Experiences	Defended Sep 2023
	Olin College of Engineering , Needham, MA B.S., Engineering, Robotics	May 2017
INDUSTRY EXPERIENCES	Google Brain, Magenta Mountain View, CA <i>Research Intern; hosted by Jesse Engel and Anna Huang</i>	Jun - Sep 2021
	Google Brain, People AI Research Mountain View, CA <i>Research Intern; hosted by Carrie J Cai</i>	Jun - Sep 2019
	Babson Summer Venture Program , Boston, MA <i>Co-founder of CrossRoad Optics, Inc.</i>	Jun - Aug 2017
	Uber ATC , Pittsburgh, PA <i>Software Engineer Intern</i>	May - Aug 2016
	Indico Enterprise AI , Boston, MA <i>Machine Learning Research Development Intern</i>	Jun - Aug 2015
	DuPont Pioneer , Boston, MA <i>Data Science Intern</i>	May - Aug 2014
AWARDS & HONORS	Recipient of Google PhD Fellowship in Human Computer Interaction , 2022.	
	Finalist for Meta PhD Research Fellowship in AR/VR Human Computer Interaction, 2022.	
	Design Research Fellow, Segal Design Research Cluster at Northwestern, Funding support from 2018 - 2019.	
	Honorable Mention, NSF Graduate Research Fellowship, 2017.	

GRANTS/FUNDING	Co-PI - Stanford Psychiatry Innovator Grant. \$25,000. Dec 2024 <i>CARE: A Large Language Model-powered platform for Deliberate Practice of Psychotherapy Skills</i>
	Co-PI - Stanford Human-AI (HAI) Seed Grant. \$75,000. Nov 2024 <i>Develop and deploy LLM-based Training Tools in Psychotherapy Learning Communities.</i>
WORKING PAPERS	Ryan Louie , Raj Sanjay Shah, Ifdita Hasan Orney, Juan Pablo Pacheco, Emma Brunskill, Diyi Yang. Can LLM-Simulated Practice and Feedback Upskill Human Counselors? A Randomized Study with 90+ Novice Counselors <i>Pre-print and Under review CHI'26</i>
	Ryan Louie , Ellen Converse, Emma Brunskill, Diyi Yang. Micro-Randomized Trial of an AI-Simulated Practice Tool for Therapeutic Skills <i>Under review CHI'26</i>
	Ryan Louie* , Ellen Converse*, Debra Safer, Juan Pablo Pacheco, William Fang, Jamie Kent, Bruce Arnow, Emma Brunskill, Diyi Yang. AI-Augmented Practice in Psychotherapy Education: A 16-Week Comparative Classroom Deployment Study <i>Under review JMIR</i>
CONFERENCE & JOURNAL PAPERS	Dora Zhao, Qianou Ma, Xinran Zhao, Chenglei Si, Chenyang Yang, Ryan Louie , Ehud Reiter, Diyi Yang, and Tongshuang Wu. SPHERE: An Evaluation Card for Human-AI Systems In <i>Findings of the Association for Computational Linguistics: ACL 2025</i>
	Han Zhang, Jiaqi Zhang, Yuxiang Zhou, Ryan Louie , Taewook Kim, Qingyu Guo, Shuailin Li, Zhenhui Peng. MentalImager: Exploring Generative Images for Assisting Support-Seekers' Self-Disclosure in Online Mental Health Communities. <i>To appear at CSCW 2026.</i>
	Ryan Louie . Ananjan Nandi, William Fang, Cheng Chang, Emma Brunskill, Diyi Yang. Roleplay-doh: Enabling Domain-Experts to Create LLM-simulated Patients via Eliciting and Adhering to Principles. In <i>Proceedings of The 2024 Conference on Empirical Methods in Natural Language Processing</i>
	Alicja Chaszczewicz, Raj Shah, Ryan Louie , Bruce Arnow, Robert Kraut, Diyi Yang. Multi-Level Feedback Generation with Large Language Models for Empowering Novice Peer Counselors. In <i>Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i>
	Ryan Louie , Darren Gergle, Haoqi Zhang. Affinder: Expressing Concepts of Situations that Afford Activities using Context-Detectors. In <i>Proceedings of ACM CHI Conference on Human Factors in Computing Systems (CHI 2022), 2022.</i>
	Ryan Louie , Jesse Engel, Anna Huang. Expressive Communication: Evaluating Developments in Generative Models and Steering Interfaces for

Music Creation. In *Proceedings of International Conference on Intelligent User Interfaces (IUI 2022)*, 2022.

Ryan Louie, Kapil Garg, Jennie Werner, Allison Sun, Darren Gergle, Haoqi Zhang. **Opportunistic Collective Experiences: Identifying Shared Situations and Structuring Shared Activities at Distance.** In *Proceedings of ACM Human-Computer Interaction (CSCW 2020)*, 2020.

Ryan Louie, Andy Coenen, Cheng Zhi Huang, Michael Terry, Carrie J Cai. **Novice-AI Music Co-Creation via AI-Steering Tools for Deep Generative Models.** In *Proceedings of ACM CHI Conference on Human Factors in Computing Systems (CHI 2020)*, 2020.

Chris Yoon, **Ryan Louie**, Jeremy Ryan, MinhKhang Vu, Hyegi Bang, William Derksen, Paul Ruvolo. **Leveraging Augmented Reality to Create Apps for People with Visual Disabilities: A Case Study in Indoor Navigation.** In *The 21st International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2019)*, 2019. **Best Artifact Award.**

SHORT PAPERS

Onno P Kampman, Michael Xing, Charmaine Lim, Ahmad Ishqi Jabir, **Ryan Louie**, Jimmy Lee, Robert JT Morris. **Conversational Self-Play for Discovering and Understanding Psychotherapy Approaches** In *AI4X 2025, Singapore*.

Nic Becker, **Ryan Louie**, John Thickstun, Percy Liang. 2024. **Designing Live Human-AI Collaboration for Musical Improvisation** In *CHI Workshop on Generative AI and HCI (GenAICHI) 2024*

Paul Ruvolo, **Ryan Louie**, and Eric Jerman. 2023. **The Cane Game: An Educational Tool for Orientation and Mobility.** In *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems (CHI 2023)*

Ryan Louie, Taewook Kim, Zhenhui Peng. 2023. **Rawal: A Reflection Assistant with Active Listening for Peer Support Chats.** In *CHI 2023 Workshop on Integrating Individual and Social Contexts into Self-Reflection Technologies*

Ryan Louie, Jesse Engel, Anna Huang. **A Unified Evaluation of Expressive Generative Models and Steerable Interfaces for Music Creation.** In *GenAICHI Workshop at CHI 2022. May 2022. [Workshop Info]*

Nina Cong, Kevin Cheng, Haoqi Zhang, **Ryan Louie**. **Collective Narrative: Scaffolding Community Storytelling through Context-Awareness.** In *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing (CSCW'21 Companion)*, 2021.

Ryan Louie, Andy Coenen, Cheng Zhi Huang, Michael Terry, Carrie J Cai. **Cococo: AI-Steering Tools for Music Novices Co-Creating with Generative Models** In *IUI 2020 workshop on Human-AI Co-Creation with Generative Models. April 2020.*

MENTORED STUDENT RESEARCH COMPETITION PAPERS	Cindy Hu. Self-Disclosure for Early Relationship Development through Situated Prompts in Opportunistic Collective Experiences . In <i>Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems (CHI 2022)</i> , 2022. 2nd Place in Undergraduate Student Research Competition.		
	Gabriel Caniglia. Cast: A Context-Aware Collaborative Storytelling Platform . In <i>Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems (CHI 2020)</i> , 2020.		
	Jennie Werner, Allison Sun. Cerebro: A Platform for Opportunistic Collective Experiences . In <i>Extended Abstracts ACM CHI Conference on Human Factors in Computing Systems (CHI 2018)</i> , 2018. 2nd Place in Undergraduate Student Research Competition.		
INVITED-LECTURES	<i>CARE: Training Novice Counselors via LLMs</i> . AI and Mental Health Course, University of Virginia, Oct 7th 2025.		
	<i>CARE: Training Novice Counselors via LLMs</i> . Dagstuhl Seminar for NLP in Mental Health, Germany, Sept 2025.		
	<i>CARE: Training Novice Counselors via LLMs</i> . Stanford University NLP Group, July 2025.		
	<i>Scaling Counseling Support with CARE: an AI-Assisted Training System Co-Designed with Mental Health Experts</i> . Stanford University CS329X Human-Centered LLMs , Oct 2024.		
	<i>Scaling Counseling Support with CARE: an AI-Assisted Training System Co-Designed with Mental Health Experts</i> . Seminar in Human Computer Interaction at Stanford University Computer Science, July 2024.		
	<i>Human AI Interfaces: Realizing AI's Potential to Foster Desired Human Experiences</i> . Seminar in Informatics School at New Jersey Institute of Technology, Feb 2023.		
	<i>Empowering Novice Composers to Realize their Creative Goals using Generative Models</i> . In Music + AI reading group at the Montreal Institute of Learning Algorithms (MILA) + Vector Institute, April 2022. [Talk]		
TEACHING EXPERIENCE	Stanford University , Stanford, CA USA		
	<i>Project mentor</i> , CS329X Human-Centered LLMs		Fall 2025
	<i>Project mentor</i> , CS224N NLP with Deep Learning		Winter 2024
	Northwestern University , Evanston, IL USA		
	TA, CS315 Design Technology Research		Winter & Spring 2022
	<i>Project Mentor</i> , CS315 Design Technology Research		Fall 2017 - 2023
	TA, CS314/COMM351 Technology Human Interaction		Winter 2021

	Olin College of Engineering , Needham, MA USA <i>Teaching Assistant</i> , ENGR3590 Computational Robotics Spring 2017 <i>Teaching Assistant</i> , ENGR/MTH2199C Data Science Spring 2015 <i>Teaching Assistant</i> , MTH2220 Linearity II Fall 2014 <i>Teaching Assistant</i> , MTH2210 Linearity I Spring 2014
EXTERNAL SERVICE	Bay Area Social Co-Organizer, <i>Society for Digital Mental Health</i> Jun-Oct 2025 <i>Under the leadership of Ellie Burgess PhD, we hosted two events (30+ attendees) in Mountain View, CA and San Francisco, CA for researchers, clinicians, startup founders, industry researchers to discuss technology/AI in mental health.</i> Reviewer for HCI community (CHI'26, CSCW'26, CHI'25, CHI'24, CHI'23, CSCW'22, DIS'22) Reviewer for NLP community (ACL SRW'25) Program committee: IUI 2022 Workshop on Human-AI Co-Creation with Generative Models.
INTERNAL SERVICE	Stanford NLP CLIPs Mentor 2024-2025 <i>Facilitate Screening of Undergraduate Applicants to Stanford NLP group</i> Social Organizer, Stanford NLP Group Retreat Winter 2024 <i>Organized a Research Karoke Event</i> Social Organizer, Stanford NLP Group Retreat Winter 2024 Moderator for a Panel on Industry Careers for PhDs in HCI and Communication Studies, Spring 2021. Poster and Demos Organizer, Northwestern Technology and Social Behavior Prospective Student Visit Day, Winter 2019.
BLOGS AND ONLINE ARTICLES	<i>Using AI to Train Peer Counselors</i> , Stanford University Human-Centered AI Blog <i>Supporting Designers in Expressing Human Experiences to AI and Context-Aware Technologies</i> , Northwestern Computer Science Website <i>HCI and ML: Putting People First</i> , Google Magenta Team Website <i>R. Louie. Neural Image Captioning for Mortals. <i>indico Blog: Resources for exploring machine learning and data science</i>, 2015.</i>
PATENTS / PRIOR ART	Ryan Louie , Nicole Rifkin, Brandon Rohrer. Method for quantifying crop health using overhead imagery . <i>The IP.com Prior Art Database</i> , DuPont Pioneer, 2015.

EXTRACURRICULAR
ACTIVITIES

Northwestern University Table Tennis Club *President* Fall 2019 - 2023
Competitor in National Collegiate Table Tennis Association Tournaments Fall
2017 - Now

- During my 4 years as president, I led 12 - 16 students per year to represent Northwestern to compete against other colleges and universities across Varsity Co-Ed, Womens, and JV Co-Ed divisions.
- Qualified and Competed at Nationals Competition (top 24 schools in North America) in 2018, 2022, and 2023 held in Round Rock, TX.